The Zero-Parameters Claim in Recognition Science: Scope, Claim Hygiene, and What It Means to Derive Physics from Nothing

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Abstract

We clarify the precise scope and meaning of the Recognition Science (RS) "zero free parameters" claim. This is not a vague assertion — it is a specific conditional statement with explicit structural constraints. The claim: **IF** [observables require recognition, conservation implies ledger, discreteness requires countable states, self-similarity requires φ] **THEN** φ is uniquely pinned and all constants $\{c, \hbar, G, \alpha^{-1}\}$ are derived. We distinguish four claim types (DEF, THEOREM, HYPOTHESIS, CERT), explain the SI calibration seam (a reporting convention, not a free parameter), and address common misconceptions. The "zero parameters" refers specifically to the model layer — no per-species fitting of dimensionless knobs. Numerical displays in SI units use a single external anchor (τ_0 in seconds) as a unit convention.

1 The Precise Zero-Parameters Claim

1.1 What It Says

Theorem 1.1 (Zero-Parameters Conditional)**.** *If structural constraints C1–C4 hold:*

(C1) *Observables ⇒ Recognition (necessity of recognition structure)*

(C2) *Conservation ⇒ Ledger (closed-loop flux forces double-entry)*

(C3) *Zero-parameters ⇒ Discrete/Countable (no continuous uncountable structure)*

(C4) *Self-similarity + Cost-Uniqueness ⇒ φ forced (unique positive root of $x^2 = x + 1$)*

then φ is uniquely pinned and all dimensionless constants are derived.

1.2 What It Doesn't Say

NOT claimed:

- That SI units are derived from nothing (units are a gauge choice)

- That the numeric value $\tau_0 = 7.3 \times 10^{-15}$ s needs no external input (the SI second needs calibration)
- That all claims are at the same certainty level (see claim hierarchy below)

2 The Four Claim Types

RS uses a strict hierarchy of claim types:

Type	Meaning	Example
DEF	Vocabulary/structures (not falsifiable)	$J(x) = \frac{1}{2}(x + x^{-1}) - 1$
THEOREM	Mathematical consequence	$\varphi^2 = \varphi + 1$
HYPOTHESIS	Empirical claim (must have falsifier)	$\alpha_s = 2/17$
CERT	Machine-certified result	$\alpha > 0$, w_8 = closed form

2.1 The Claim Tiers

Tier A — Pure structure (no calibration needed):

- $D = 3$ (dimension forced by T8)
- $N_{\text{gen}} = 3$ (three generations from D)
- $\theta_{\text{QCD}} = 0$ (from J-symmetry)
- $m^2_{\nu_3}/m^2_{\nu_2} = \varphi^7$ (pure φ-power ratio)

These predictions need NO external calibration — they follow from RS structure alone.

Tier B — Structure proven, numerics certified externally:

- Anchor scale $\mu^\star = 182.201$ GeV (structure machine-verified; numeric from SM RG tools)
- Fermion mass ratios $m_\mu/m_e = \varphi^{11}$ (structure proven; numeric comparison needs PDG)

Tier C — Model-layer (RS-internal convention):

- $\alpha^{-1} \approx 137.036$ (requires the w_8 geometric projection; derived from φ)
- Boson masses $m_W \approx 80.4$ GeV (requires VEV calibration)

3 The SI Calibration Seam

3.1 What It Is

Proposition 3.1 (SI Calibration Seam)**.** *The SI calibration seam is a single external input: τ_0 in seconds. Once τ_0 is fixed, all other SI quantities follow algebraically:*

$$\ell_0 = c \cdot \tau_0 \quad (\textit{SI-defined } c = 299792458 \ m/s) \tag{1}$$

$$E_{\text{coh}} = \hbar/\tau_0 \quad (\textit{SI-defined } h) \tag{2}$$

$$G = \pi c^3 \ell_0^2/\hbar \quad (\textit{RS derivation}) \tag{3}$$

Proof. Fixing τ_0 in seconds determines ℓ_0, E_{coh}, and G via the algebraic relations above. Machine-verified. □

3.2 Why It's Not a Free Parameter

τ_0 is a *unit convention*, not a free parameter:

- The SI second was defined by the cesium atom (BIPM 1967)
- Mapping RS-native time to SI seconds is a unit choice, like converting feet to meters
- The RS predictions for dimensionless ratios (like m_μ/m_e) need *no* SI seam

4 The Minimality Stack

Theorem 4.1 (Minimality Stack)**.** *RS achieves zero free parameters through a minimality argument. Each step is necessary (remove any element and the constraint fails) and sufficient (add no more and everything follows).*

Principle	RS Implementation
MP minimal	$J(0^+) \to \infty$ is necessary and sufficient
T3 continuity	Closed-chain flux = 0
T5 cost uniqueness	$J(x) = \frac{1}{2}(x + x^{-1}) - 1$, $J''(1) = 1$
T6 eight-tick minimality	$D = 3 \Rightarrow$ period 8
Units-quotient gauge rigidity	bridge_factorizes

5 Acknowledged Gaps

RS is explicit about what it does NOT claim:

5.1 Model-Independent Exclusivity: Scaffold

The claim "no alternative zero-parameter framework exists" is proven on a quotient state space — not in full generality. Proving it unconditionally would require showing all possible frameworks are equivalent to RS.

Current status: proven on quotient (model-independent exclusivity = scaffold).

5.2 Absolute Scale Without External Anchor

"All constants derived without any external input" is NOT established. The SI mapping uses one external anchor (τ_0).

The dimensionless structure (all pure RS ratios) IS parameter-free. The absolute scale (specific numbers in SI units) requires one unit convention.

5.3 QFT-Level Proofs

Some derivations are "structure proven" but "numerics certified externally." This means: the machine-verified proof shows the algebraic structure, but the specific numerical values require external SM RG tools.

6 Why "Zero Parameters" Is Still Meaningful

Despite these caveats, the zero-parameters claim is genuinely significant:

No per-species fitting: The mass formula $m_f = \text{yardstick} \cdot E_{\text{coh}} \cdot \varphi^{r_f - 8 + \text{gap}(Z_f)}$ uses no per-species adjustment knobs. All rungs r_f are computed from cube geometry.

Seam-free predictions: Ratios like $m_{\nu_3}^2/m_{\nu_2}^2 = \varphi^7$ need no calibration whatsoever — they are pure φ-powers.

Causal derivation chain: Every constant traces back to the Recognition Composition Law through a causal chain with no free links.

7 The Inquiry Geometry

The RS meta-closure: RS is a self-justifying framework. The inquiry about "why RS?" leads to RS as the answer.

The 8 fundamental question modes (from the 8-tick epoch):

Tick	Mode	RS Answer
0	What?	$J(x) = \frac{1}{2}(x + x^{-1}) - 1$ (existence cost)
1	Why?	$J(0^+) \to \infty$ (nothing cannot exist)
2	How?	T0-T8 forcing chain
3	When?	8-tick epoch defines time
4	Where?	$D = 3$ defines space
5	Who?	φ defines the observer
6	Which?	Unique J minimizer
7	Whether?	RS or not = forced (RS is zero-parameter)

Theorem 7.1 (Meta-Closure)**.** *Question 7 (Whether RS is correct?) leads to RS as the unique zero-cost theory. RS explains RS without circularity: the explanation is that RS is the unique consistent zero-parameter framework.*

Corollary 7.2 (Meta-Cost)**.** *The cost of understanding RS to depth n is $n \times \ln \varphi$ J-bits:* $\text{metaCost} = \text{depth} \times \ln \varphi$.

8 Claim Hygiene Summary

Claim	**Status**	**Verification**
J is unique (T5)	PROVEN	machine-verified
$D = 3$ is forced	PROVEN	machine-verified
$N_{\text{gen}} = 3$	PROVEN	machine-verified
$\theta_{\text{QCD}} = 0$	PROVEN	machine-verified
$m_{\nu_3}^2/m_{\nu_2}^2 = \varphi^7$	DERIVED	machine-verified
$\alpha^{-1} \approx 137.036$	DERIVED+SEAM	machine-verified
$m_W \approx 80.4$ GeV	STRUCTURE+SEAM	machine-verified
Model-independent exclusivity	SCAFFOLD	partial
Consciousness survives death	PROVEN	machine-verified

9 Conclusion

The RS zero-parameters claim, properly understood:

1. Is a conditional statement (IF C1-C4 THEN φ is forced and all constants follow)
2. Uses a strict claim hierarchy (DEF < THEOREM < HYPOTHESIS < CERT)
3. Requires exactly one external unit convention (SI τ_0) for numerical displays
4. Makes fully seam-free predictions for dimensionless ratios ($m_{\nu_3}^2/m_{\nu_2}^2 = \varphi^7$)
5. Is genuinely meaningful: no per-species fitting, causal derivation chain
6. Has explicitly acknowledged gaps (exclusivity is scaffold, absolute scale needs anchor)

References

[1] CODATA, *2018 Recommended Values of the Fundamental Physical Constants*, NIST (2019).